

Comprehensive Project & Procedural Document

SYSTEM ARCHITECTURE & WORKSTATION CONTROL SURFACE REGISTRY

Host Node:	corsair-one	OS Environment:	Kubuntu 24.04 LTS (Plasma 5.27/X11 Baseline)
Target Tier:	Deterministic Systems Control	Authority:	Control-Surface Registry v2

1. EXECUTIVE SUMMARY

This document consolidates all authoritative project configurations, operational metadata, and architectural constraints for the `corsair-one` workstation into a unified, structured deployment record. By centralizing the system topology—ranging from physical Stratum 0 hardware parameters up to Stratum 17 configuration encapsulation layouts—this ledger eliminates previous formatting discrepancies and establishes absolute, reproducible control over the workstation stack. Future initiatives, including downstream image processing pipelines, remain explicitly segregated from this system administration baseline.

2. DOCUMENT REVISION PROCEDURES

To preserve the integrity of the workstation's configuration state and ensure natural, idiomatic technical transmission, all future system modifications and structural data logging must conform strictly to the following processing loop:

- Drafting Phase:** Raw system data, command output logs, and environment configurations are submitted directly through the LLM canvas interface to initialize an integrated operational state.
- Refining Phase:** The system leverages rigorous, explicit direct prompting to sanitize superficial jargon, resolve formatting collisions, and align statements with physical system causal realities.
- Exporting Phase:** Following validation, the unified record is compiled natively into a structured PDF document within the chat execution environment for immediate host integration and offline tracking.

3. EXHAUSTIVE DATA COMPILATION

The matrix below delineates the absolute ground truth of the system state, categorized by the core operational strata governing the workstation surface.

3.1 Stratum 0: Physical Hardware Base

COMPONENT	HARDWARE SPECIFICATION & CONFIGURATION POSTURE
Motherboard	ASUS TUF X299 Mark 1 (Intel X299 Chipset, LGA 2066 Socket)
CPU	Intel Core i9-10900X (Cascade Lake-X, 10C/20T, 3.70 GHz Base / 4.70 GHz Turbo, 48 PCIe Lanes, AVX-512 Dual FMA Units, 165W TDP)
RAM	160 GB DDR4 Quad-Channel (8 Slots). Active: JEDEC Default DDR4-2400. Channels A1–B2: 32 GB modules (2666 MHz native); Channels C1–D2: G.Skill 8 GB modules (2133 MHz rated).
GPUs	Dual NVIDIA TITAN Xp Collectors Edition (Pascal, GP102, sm_61, 12 GB GDDR5X each, PCIe 3.0 x16/x16, Non-NVLink, No Tensor Cores).
Storage Array	Four-drive architecture: sda (232 GB legacy LVM), sdb (1.86 TB Seagate, ~84% cap), sdc (2.79 TB LUKS-encrypted, unmounted), sdd (1.86 TB Active OS Drive).
PCIe Topology	PCIEX16_1 at x16; PCIEX16_2 at x16; PCIEX16_3 at x8; PCIEX4_1 forced to x1; PCIEX4_2 at x4. M.2_1 shares explicit bandwidth with PCIEX4_2 and SATA ports 5–8.
Peripherals	Native Instruments Komplete Audio 6 (USB Audio), Dual Intel NICs (enp2s0/igb active; eno1/e1000e down), NZXT Kraken2 AIO Liquid Cooler.

3.2 Stratum 1 & 5: Kernel, Driver, and Compute Constraints

The terminal nature of the Pascal hardware architecture imposes rigid boundaries that propagate through the entire software stack. These constraints are non-negotiable and treated as physical laws.

DOMAIN	BOUNDARY LAYER & ENFORCEMENT RULE
NVIDIA Driver	Branch 580.159.03 live. Driver 590+ drops Pascal support entirely. Locked permanently via multi-layer pinning.
Apt Pinning State	<code>/etc/apt/preferences.d/nvidia-580-pin</code> set to Pin-Priority 1001 for 580* targets; priority -1 for non-580 builds. Hard locks (<code>apt-mark hold</code>) active across 17 packages.
CUDA Toolkit	Absolute ceiling at 12.9.2 . CUDA 13 completely deprecates Pascal architectures.
cuDNN Library	Absolute ceiling at 9.11.1 . Version 9.12.0 drops sm_61 execution paths.
Host Kernel	Hard limit at \$!e\$ 6.17 . Upstream HWE Kernel 7.0 (arriving August 2026) breaks Driver 580 DKMS module compilation.
Python Runtime	Strict compliance with PEP 668. Banned use of <code>--break-system-packages</code> . Execution strictly via isolated <code>uv + venv</code> targets under <code>~/envs/</code> .

3.3 Stratum 17: Authoritative Configuration Backup Architecture

Deterministic reconstruction of the workstation relies on versioning the discrete layout files detailed below. Any configuration adjustments must target these precise file pathways using native KDE scripting tooling.

- `~/.config/*rc` — Absolute state ledgers for all individual KDE Plasma subsystems.
- `~/.config/plasma-workspace/` — Persistent desktop session layouts, applet position entries, and geometry trees.
- `~/.local/share/plasma/` — Custom workspace look-and-feel components, configurations, and internal theme assets.
- `~/.local/share/applications/` — Custom web-application launcher definitions and isolated `.desktop` entry files.
- `/etc/apt/preferences.d/nvidia-580-pin` — Underlying driver shielding script protecting against upstream repo contamination.
- `/etc/modprobe.d/` — Kernel runtime configurations, including mandatory nouveau blacklisting profiles.
- `/etc/sddm.conf.d/` — Display manager configurations defining DM handoff constraints and initial authorization layout.

4. STANDARD COMMAND SAFETY PROTOCOL

Every structural command execution block target on the host must be subjected to validation before run-time execution. Commands must be separated by purpose, isolating read-only data discovery sequences from active mutable transformations. Joining must rely on semicolons (;) rather than standard conditional conjunctions to guarantee predictable fault tracking.